Athena-SDA: quantitative orbital noise detection and micro-anomaly analysis for military-interest satellites using past-only Isolation Forest and walk-forward validation against open-source report anchors

Athena-SDA Project
`https://github.com/CostaJr007/Athena-SDA`
IBM SkillsBuild AI Builders — *Advance Space Exploration with AI*
Mathematical foundation and experimental validation (Claims A+B)

August 2026

Abstract

We present **Athena-SDA**, a military-first Space Domain Awareness (SDA) system for **quantitative orbital noise analysis**, **micro-anomaly detection**, and **operational alerts** on platforms of interest (*suspects*), using public TLEs and GFZ space weather. We build a **quantitative feature vector** (Shannon entropy, multi-scale Hurst exponent, CUSUM, Kolmogorov proxy, stationarity tests, F10.7/Ap/Kp space weather, among others) and train an **Isolation Forest** on normality anchors (civil *baseline* and protected *assets*) under a **past-only** protocol. We validate with **walk-forward** windows aligned to **open-source reports** and civil EO **placebo** controls. On the GEO panel (Luch/Olymp-K, Shiyan-12, Luch-2): **5/5** hard hits at threshold 0.50 with mean max score ≈ 0.65, versus **0/7** hard hits and mean ≈ 0.46 for civil EO placebos (gap ≈ 0.19; one-sided Mann–Whitney $p \approx 0.001$). A priority layer (XGBoost weak labels, suspect$\times$asset pairs, fuzzy rules) and the Bob copilot explain and rank attention without changing the quantitative score. Short definitions of each mathematical tool are collected in the **Glossary** (Appendix A).

Contents

1 Introduction 2

2 Military-first doctrine 3

3 Data 3
3.1 Orbital series (TLE) . . . 3
3.2 Space weather . . . 3
3.3 Open-source anchors . . . 3

4 Quantitative noise vector 3
4.1 Shannon entropy . . . 3
4.2 Kolmogorov complexity proxy . . . 3
4.3 Multi-scale Hurst (R/S) . . . 4
4.4 Kernel L_1-CUSUM . . . 4
4.5 Mandelbrot / Hill tail . . . 4
4.6 ADF . . . 4
4.7 Deltas and maneuvers . . . 4
4.8 Topology and geometry (proxies) . . . 4
4.9 Space weather . . . 4
4.10 Excluded from IF . . . 4

5 Past-only Isolation Forest 4

6 Walk-forward protocol 5

7 Priority layer 5

8 Experimental design 5

9 Results 5
9.1 Claims A+B — GEO panel 5
9.2 Per-event sample 6
9.3 Score curves 6
9.4 Expanded panel 9

10 Discussion 9

11 Limitations 9

12 Reproducibility 9

13 Conclusion 9

A Glossary of mathematical and model tools 9

---

1 Introduction

SDA operators cannot monitor the full catalog with equal attention. Athena-SDA focuses on **quantitative analysis of orbital time series** and **detection of noise / micro-anomalies** on military-interest objects, producing **alerts** and attention priority relative to platforms to protect.

The system combines:

1. public data (multi-year TLEs + space weather);
2. a deterministic quantitative noise framework (Sec. 4);
3. past-only Isolation Forest under military-first doctrine (Sec. 5);
4. walk-forward validation against report anchors and placebos (Sec. 6–9);
5. priority and explanation layers (Sec. 7).

Research question. Does a quant + past-only Isolation Forest pipeline, trained on *baseline+asset* normality, separate military-interest windows aligned with open-source reports from civil EO controls under the same protocol?

Formal claims.

Claim A.
Interest cases (open-source anchors) show elevated anomaly scores and/or hard hits near t_{peak}.

Claim B.
Civil EO placebos under the same protocol show lower scores and near-zero hard-hit rate at $\mathrm{thr} = 0.50$.

2 Military-first doctrine

Role	Examples	ML use
asset	ISS, GPS, DMSP, allied SAR	IF normality anchor; protect / pair target
suspect	Luch/Olymp-K, Yaogan, Shiyan, CSS	Primary micro-anomaly / noise detection
baseline	TERRA, AQUA, Landsat, NOAA	IF training (stable orbit reference)

Suspects are excluded from IF normality training. Commercial mega-constellations (e.g. Starlink) are excluded from that training set. Analysis targets the military watchlist and quant pipeline; the 3D UI is for demonstration.

3 Data

3.1 Orbital series (TLE)

- Filtered history for 24 NORADs (∼2014-01-01 to 2026-07-27), ∼**249,580** epochs.
- Sources: public TLE history mirror + daily CelesTrak GP ingest.
- Elements: semi-major axis (SMA), eccentricity, inclination, RAAN, mean motion, timestamps.

3.2 Space weather

GFZ Potsdam (Kp, ap/Ap, observed/adjusted F10.7, sunspots):

- $F_{10.7}$, $F_{10.7}^{\mathrm{adj}}$, Ap, $\overline{K_p}$, sunspot number;
- 7-day means and deltas;
- `geomagnetic_storm`$= 1$ if $\mathrm{Ap}_{\mathrm{max,7d}} \geq 30$.

Space weather contextualizes atmospheric drag (especially LEO) relative to SMA changes.

3.3 Open-source anchors

Events in `events_walkforward.json` with $(t_{\mathrm{start}}, t_{\mathrm{peak}}, t_{\mathrm{end}})$ and public sources (Gunter, CSIS, SWF, press, dual-use catalog notes). Examples: Luch/Olymp-K near Intelsat/Athena-Fidus; Shiyan-12; Luch-2; CSS Tianhe; placebos TERRA, AQUA, Landsat, NOAA.

4 Quantitative noise vector

Let $a(t)$ be the SMA series (km) on a window of $n \geq 20$ epochs. `src/engine.py` produces $x \in \mathbb{R}^d$ for the Isolation Forest. (See Appendix A for short tool definitions.)

4.1 Shannon entropy

Differences $\Delta a_i = a_{i+1} - a_i$, histogram p_k:

$$H = -\sum_{k:\, p_k > 0} p_k \log_2 p_k. \tag{1}$$

4.2 Kolmogorov complexity proxy

Tokens $\{U, D, S\}$ on Δa and zlib compression:

$$K_{\mathrm{proxy}} = \mathrm{clip}\left(\frac{\ell_{\mathrm{comp}}}{\ell_{\mathrm{raw}}},\, 0,\, 1\right). \tag{2}$$

4.3 Multi-scale Hurst (R/S)

H on the full window and a short recent tip; gap

$$\Delta H = H_{\text{full}} - H_{\text{short}}. \tag{3}$$

4.4 Kernel L_1-CUSUM

Accumulated deviations from local median / MAD, mapped to $[0, 1]$.

4.5 Mandelbrot / Hill tail

Extreme-value score above a high quantile of the series.

4.6 ADF

Augmented Dickey–Fuller p-value on $a(t)$.

4.7 Deltas and maneuvers

ΔSMA over 7d/30d and CUSUM-peak maneuver counts.

4.8 Topology and geometry (proxies)

H0/H1 (proxy pairwise-distance mode), Chern–Simons proxy on $h = r \times v$, Ollivier–Ricci proxy via local Wasserstein. Reconstructed from orbital elements (not full SGP4 ephemerides).

4.9 Space weather

$F_{10.7}$, Ap, Kp and 7-day derivatives in the same vector x.

4.10 Excluded from IF

Distance to assets and cointegration feed the priority layer only.

5 Past-only Isolation Forest

$$s(x) = \text{clip}\big(0.5 - \text{decision\_function}(x),\ 0,\ 1\big). \tag{4}$$

Training. Windows with window_end < cutoff; NORADs **baseline + asset**; contamination ~ 0.06–0.08; `random_state=42`; separate monitor and pipeline artifacts (`registry.json`).

Threshold.

$$thr = \max\big(0.50,\ p_{95}(s_{\text{normality}})\big). \tag{5}$$

Primary tables use thr $= 0.50$.

Hard hit. Any fold with $s \geq 0.50$ inside $[t_{\text{peak}} \pm 45\,\text{d}]$.

6 Walk-forward protocol

For each event ($t_{\text{start}}, t_{\text{peak}}, t_{\text{end}}$):

1. `asof` grid with 14-day step.
2. At each `asof`: fit IF only on normality windows ending before `asof`−3 days.
3. Score the target NORAD on the current window.
4. Metrics: hard hit, max s, pre-peak mean, `noise_ramp`, `first_fold_hit`.

Locked plan: `PROTOCOL_PREREGISTRATION.md`.

7 Priority layer

Complements noise detection for operator attention ranking:

- XGBoost with weak labels (operational tiers);
- suspect×asset distance and cointegration (`pair_risk`);
- Mamdani fuzzy rules and Kelly-style attention sizing;
- Bob: natural-language briefing from computed scores.

Claims A+B use only the walk-forward IF score.

8 Experimental design

GEO panel (primary). Interest: 5 events (Luch-1 ×3, SY-12, Luch-2), 3 unique NORADs. Civil EO placebos: 7 windows (TERRA, AQUA, Landsat, NOAA), 5 unique NORADs.

Expanded panel. Up to 9 unique interest NORADs (LEO/MEO included); reported separately.

Separation test. One-sided Mann–Whitney U (interest > placebo), emphasizing means and gap.

9 Results

9.1 Claims A+B — GEO panel

Table 1: Primary results (thr= 0.50, past-only). Source: `paper_validation_latest.json`.

Group	n events	Hard hit	Mean max s	Mean pre-peak
GEO interest	5	**5/5 (100%)**	**0.651**	**0.567**
Civil EO placebo	7	**0/7 (0%)**	0.462	0.385
Gap (interest − placebo)			**0.189**	—
Mann–Whitney (max)			$p \approx 0.0013$	—

The quant+IF pipeline separates elevated noise regimes on GEO interest cases from civil EO placebos (**Claims A and B supported** on this panel).

9.2 Per-event sample

Table 2: Per-event scores (sample).

Event	NORAD	Hard	max s	Pre-peak mean
luch1_intelsat_2015	40258	yes	0.628	0.570
luch1_intelsat_mid2015	40258	yes	0.675	0.589
luch1_athena_fidus_2018	40258	yes	0.672	0.560
sy12_geo_rpo_2021_22	50321	yes	0.623	0.529
luch2_trailing_2023	55841	yes	0.658	0.586
tianhe_css_assembly_2021	48274	yes	0.644	0.556
placebo_terra_2015	25994	no	0.481	0.395
placebo_aqua_2015	27424	no	0.474	0.429
placebo_landsat8_2018	39084	no	0.460	0.367
placebo_noaa20_2023	43013	no	0.464	0.391

9.3 Score curves

Figures 1–4 show $s(\text{asof})$, threshold 0.50, and t_{peak}.

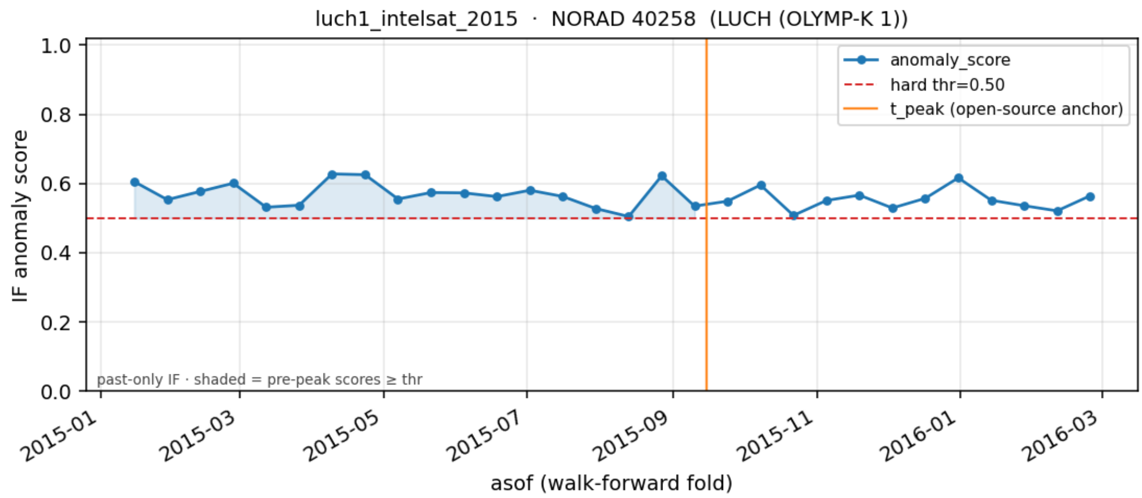
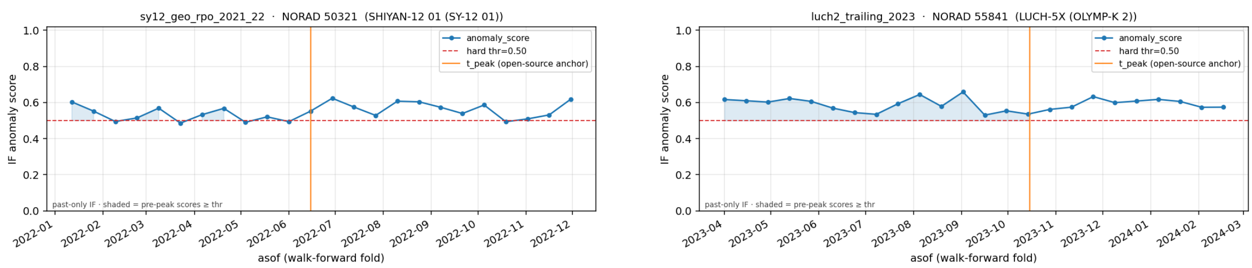
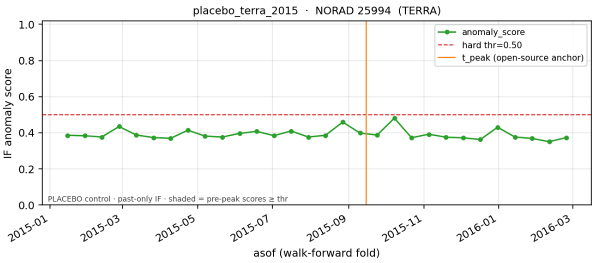


Figure 1: Luch-1 / Olymp-K (40258), Intelsat 2015 episode.

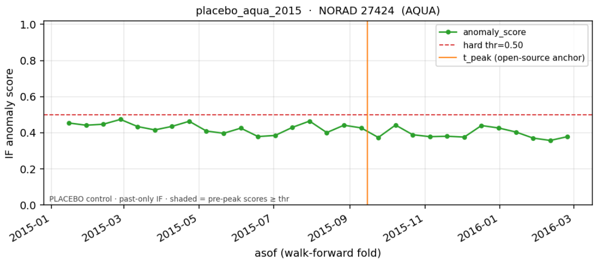


(a) Shiyan-12 (50321). (b) Luch-2 (55841).

Figure 2: Interest cases on distinct NORADs.



(a) Placebo TERRA 2015.



(b) Placebo AQUA 2015.

Figure 3: Civil EO controls.


Interest — pre-peak IF score curves
beidou3_m11_meo_2019
cosmos2550_military_leo_2022
luch1_athena_fidus_2018
luch1_geo_ops_2024
luch1_intelsat_2015
luch1_intelsat_mid2015
luch2_geo_ops_2024
luch2_trailing_2023
shiyan7_experimental_2015
sy12_geo_rpo_2021_22
sy12_interest_2024h2
tianhe_css_assembly_2021
yaogan29_recon_2020
yaogan29_sso_2025q1

9.4 Expanded panel

With 11 events / 9 unique interest NORADs, hard-hit rate $\approx 55\%$ (LEO not always ≥ 0.50), with a positive mean gap vs placebos (MW $p \approx 0.01$). The primary result of this paper is the GEO panel.

10 Discussion

Score $s(x)$ responds to elevated noise regimes on GEO interest cases aligned with open-source anchors, with clear separation from civil EO controls. Many cases show high pre-peak level and near-zero `noise_ramp`, consistent with a **persistently elevated regime** in the analysis window. Space weather contextualizes drag; pair risk and XGB rank attention when proximity to assets is present.

11 Limitations

- Small unique interest NORAD count; multi-event Luch-1 is dependent.
- Sparse, noisy TLEs; topology proxies on reconstructed geometry.
- LEO recon/dual-use hard hits less frequent at thr 0.50.
- XGB accuracy measures weak-label agreement, not Claims A+B.

12 Reproducibility

```
python scripts/run_anomaly_monitor.py train-baseline --holdout-days 1
python scripts/run_paper_validation.py --run-wf --threshold 0.50
python scripts/plot_prepeak_curves.py
cd docs/paper && pdflatex athena_sda_article.tex
```

13 Conclusion

Athena-SDA implements quantitative orbital noise analysis and micro-anomaly detection on a military-first watchlist, with past-only Isolation Forest and walk-forward validation. On the GEO panel, Claims A and B are supported (5/5 vs 0/7 hard hits; gap ≈ 0.19). The appendix glossary summarizes each mathematical tool used in the feature vector.

A Glossary of mathematical and model tools

Short definitions of **what each tool does** in Athena-SDA.

Shannon entropy
: Measures how irregular or spread out altitude steps (ΔSMA) are. High values: less predictable step patterns (irregular control / active ops).

Kolmogorov proxy
: Estimates whether the up/down/stable sequence is hard to compress. High values: more elaborate or irregular maneuver patterns.

Hurst exponent (R/S)
: Measures temporal persistence: whether deviations tend to continue in the same direction. $H > 0.5$: persistent control / micro-trajectory signature.

Multi-scale Hurst / ΔH
Compares full-window persistence with the recent tip. Separates long memory from short-tip fluctuation.

L_1-CUSUM
Detects regime breaks by accumulating deviations from a recent robust level (median/MAD). High when the series changes baseline.

Mandelbrot / Hill tail
Highlights rare extreme jumps in the altitude series (impulsive events).

ADF (Dickey–Fuller)
Tests whether the series is stationary or drifts. High p favors non-mean-reverting regimes.

ΔSMA and maneuver count
Quantify magnitude and frequency of semi-major-axis changes (relocation, intense ops).

Homology H0/H1 (proxy)
Summarizes structure of the reconstructed position cloud (connectivity / cycles across scale).

Chern–Simons (proxy)
Measures variation of specific angular momentum in reconstructed r, v — non-Keplerian deviation indicator.

Ollivier–Ricci (proxy)
Measures local neighborhood distortion in the point cloud (discrete curvature).

Space weather (F10.7, Ap, Kp, storm)
Solar and geomagnetic context. Supports interpretation of LEO ΔSMA under elevated drag.

Isolation Forest
Unsupervised model that isolates rare feature profiles. High score = atypical combination relative to normality anchors.

Past-only walk-forward
At each historical date the IF sees only the past. Scores interest and placebo event windows under that constraint.

Cointegration
Tests whether two altitude series share a common long-run trend (joint motion / escort pattern).

Suspect×asset distance
Approximate orbital proximity for attention ranking near protected platforms.

XGBoost (weak labels)
Operational priority classifier from heuristic training targets. Does not define Claims A+B hard hits.

Mamdani fuzzy
Linguistic rules with soft membership for calibration under uncertainty (e.g. stale TLE).

Kelly
Attention/resource sizing heuristic from score and severity.

Bob
Text layer that describes computed scores and context; does not recompute the IF.

References

[1] C. E. Shannon, “A mathematical theory of communication,” *Bell Syst. Tech. J.*, 1948.

[2] H. E. Hurst, “Long-term storage capacity of reservoirs,” *Trans. Amer. Soc. Civ. Eng.*, 1951.

[3] A. N. Kolmogorov, “Three approaches to the quantitative definition of information,” 1965.

[4] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation Forest,” in *ICDM*, 2008.

[5] R. F. Engle and C. W. J. Granger, “Co-integration and error correction,” *Econometrica*, 1987.

[6] B. Mandelbrot, “The variation of certain speculative prices,” *J. Business*, 1963.

[7] J. Matzka et al., Geomagnetic Kp index, GFZ Potsdam.

[8] Athena-SDA: `https://github.com/CostaJr007/Athena-SDA`, `docs/paper/*`.